

Battery Health Monitoring & RUL Prediction

Deep Learning for AUV Battery Prognostics | Project Overview & Problem Statement

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Problem Statement

- AUVs depend on batteries for propulsion, sensing, navigation, and computing; battery condition directly affects AUV endurance, reliability, and safe operation.
- Lithium-ion batteries degrade with repeated charge-discharge cycles, reducing capacity over time.
- Battery degradation is time-dependent; traditional methods struggle with complex temporal patterns.
- Deep learning models can learn these patterns directly from historical battery measurements.
- Goal: build & compare temporal deep learning models to predict SoH and RUL, with robustness and error analysis.

Key Terms

- **SoH (State of Health)**
Current battery health vs. its healthy/initial condition.
- **RUL (Remaining Useful Life)**
Useful operating life left before end-of-life.
- **Battery Degradation**
Gradual performance loss with repeated aging/use.
- **Battery Prognostics**
Estimating health now and forecasting future degradation.

System Definition (I/O)

Battery Historical Data → ***Deep Learning Model*** → ***SoH / RUL***

Inputs: cycle/history data, capacity & health indicators, operating variables.

Outputs: State of Health (SoH) and Remaining Useful Life (RUL).

Objectives, Pipeline & Modelling Approach

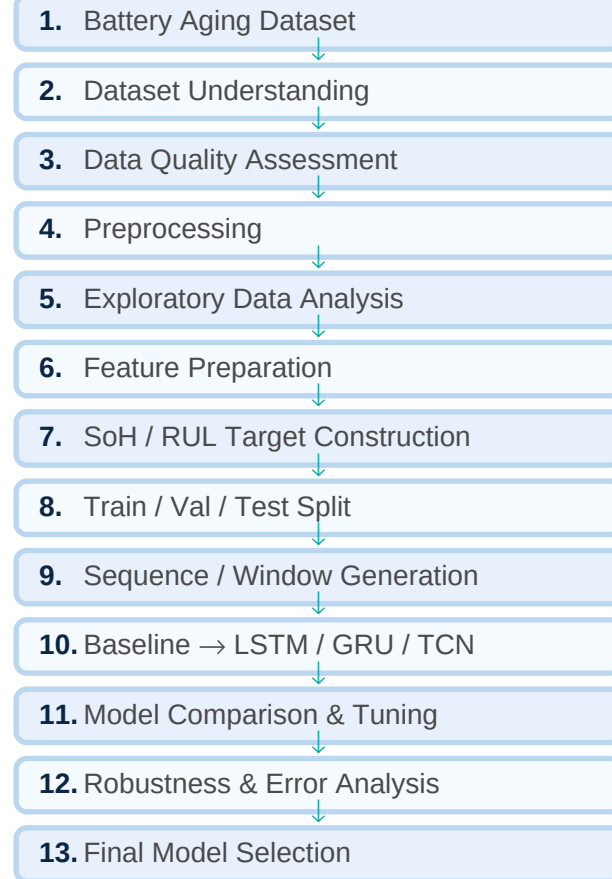
From Raw Battery Data to Deep Learning Predictions

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Core Objectives

- Understand SoH, RUL & battery prognostics fundamentals.
- Inspect the supplied battery aging dataset thoroughly.
- Build reproducible preprocessing & EDA pipeline.
- Construct proper SoH / RUL prediction targets.
- Develop a simple baseline prognostic model.
- Implement & compare LSTM, GRU, TCN (Transformer as stretch goal).
- Study robustness across cells & degradation stages.
- Analyse errors, failure cases & generalisation limits.

System Pipeline



Modelling Approach

Baseline

- Simple statistical/regression model as reference point.

LSTM

- Recurrent network to learn long-term temporal degradation trends.

GRU

- Lighter recurrent alternative, compared fairly with LSTM.

TCN

- Temporal convolutional network; compares complexity vs. performance.

Transformer

- Stretch-work model, attempted after primary models are done.

Evaluation Metrics

MAE Average absolute prediction error.

RMSE Error metric penalising larger deviations.

MAPE Percentage error relative to actual value.

R2 Variance in target explained by model.

Also tracked:

- Training time, Inference time, Model size & complexity.
- Goal: balance prediction quality vs. computational cost.

Understanding SoH & RUL:

- SoH reflects the battery's current health versus its healthy/initial condition.
- RUL estimates how much useful operating life remains before end-of-life.

Dataset & Leakage Safeguards

- Candidate datasets: NASA Li-ion Battery Aging, CALCE, or faculty-provided equivalent.
- Checked for missing/invalid/duplicate values, target coverage & temporal consistency.
- Raw dataset preserved unchanged before any preprocessing.
- Train/val/test split designed to avoid future-information leakage across cells & stages.

Data Understanding Checklist:

- Number of battery cells, cycles, and available measurements.
- Target coverage and temporal consistency across the dataset.
- Metadata review before any modelling begins.